

How Gas Helps Some Animals Float

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Passing gas may make people giggle, but for some animals it has a useful job. Gas inside their bodies can help them stay afloat.

Florida manatees mainly eat seagrass, which is tough to digest. Tiny living things called **microbes** help break it down in the animals' large guts. This process produces gas. Manatees can also swallow air.

Gas is less dense than water. Having more gas inside makes a manatee more **buoyant**, meaning it floats more easily. Its strong belly muscles and **diaphragm**, the muscle used in breathing, can squeeze gas in its intestines. Releasing some of that gas can help it dive deeper.

Sand tiger sharks use air in a different way. Many bony fish have a **swim bladder**, a gas-filled organ that helps them float. Sharks do not have **swim bladders**. Their oily livers help support them in the water, but many sharks keep swimming to avoid sinking.

Marine conservation biologist David Shiffman explains that a sand tiger shark can gulp air at the ocean's surface and use it to **hover** in the water. He says these sharks have been observed releasing air through an opening called a **cloaca**, which also handles waste and reproduction.

Some turtles use their **cloacas** for another surprising purpose. They pump water into the opening, and oxygen from that water enters their blood. This is different from releasing gas.

An everyday bodily process can reveal some clever ways animals live underwater.

Word Treasure

microbes -- Tiny living things that can help break down food.

buoyant -- Able to float easily.

diaphragm -- A muscle used in breathing.

swim bladder -- A gas-filled organ that helps many bony fish float.

hover -- Stay in one place in the water.

cloaca -- An opening in some animals used for waste and reproduction.

Background you need

Buoyancy describes how easily something floats.

Taking oxygen from water is a different process from releasing gas.

Break it down

Introduces gas as something useful for underwater animals.

Explains how manatees produce gas and use it to float or dive.

Compares swim bladders with the air used by sand tiger sharks.

Describes how some turtles take oxygen from water through their cloacas.

Why it matters

These examples help explain how animal bodies work underwater and how gas can help animals float.

Test what you remember

1. What do Florida manatees mainly eat?

- ☐ (A) Crabs
- ☐ (B) Seagrass
- ☐ (C) Sea snails
- ☐ (D) Small fish

2. What helps manatees break down tough seagrass?

- ☐ (A) Air above the water
- ☐ (B) Water around their bodies
- ☐ (C) Oil in their livers
- ☐ (D) Microbes in their guts

3. Why does gas help a manatee float?

- ☐ (A) Gas is colder than water.
- ☐ (B) Gas is saltier than water.
- ☐ (C) Gas is less dense than water.
- ☐ (D) Gas is heavier than water.

4. What can releasing some gas help a manatee do?

- ☐ (A) Swim toward shore
- ☐ (B) Dive deeper
- ☐ (C) Warm its body
- ☐ (D) Eat tougher plants

5. How does a sand tiger shark get air that helps it hover?

- ☐ (A) It gathers air from sand.
- ☐ (B) It takes air from seagrass.
- ☐ (C) It draws air from its liver.
- ☐ (D) It gulps air at the surface.

6. What enters some turtles' blood when they pump water into their cloacas?

- ☐ (A) Oxygen from the water
- ☐ (B) Oil from the water
- ☐ (C) Food from the water
- ☐ (D) Salt from the water

Answer key (try the quiz first!): 1: B 2: D 3: C 4: B 5: D 6: A

1. David Shiffman: He explains that sand tiger sharks gulp air to help them hover and have been observed releasing it through their cloacas.